

Unconditional Questions: Semantically Legitimate, Pragmatically Degraded

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Abstract Unconditional questions such as *Whether or not the party is at Emma’s place, will it be fun?* are often judged degraded, despite the acceptability of their declarative counterparts. This paper argues that their degradation is pragmatic rather than semantic. Building on inquisitive semantics and stack-based suppositional semantics, I develop a compositional account in which unconditional questions are interpreted relative to multiple hypothetical contexts corresponding to antecedent alternatives. I then propose two pragmatic constraints on their use. First, extending Theiler’s (2021) Epistemic Bridge, I argue that every antecedent alternative must sustain an inquisitive issue with respect to the consequent question. Second, when the antecedent and consequent issues are independent, unconditional questions are semantically equivalent to ordinary questions. Their use therefore invites the inference that the antecedent issue is irrelevant to resolving the consequent question, a form of relational indifference in the sense of Rawlins (2008, 2013). The analysis explains why unconditional questions improve when the antecedent issue is salient but degrade when one antecedent alternative effectively settles the consequent issue. More generally, the results provide additional support for a suppositional approach to conditionals.

1 Introduction

Unconditional constructions have received considerable attention in the semantics of conditionals and modality (Rawlins 2008, 2013). While unconditional declaratives such as (1) are generally unproblematic, their interrogative counterparts (2) often exhibit a striking degradation.¹

- (1) Whether or not the party is at Emma’s place, it will be fun.
- (2) a. #Whether or not the party is at Emma’s place, will it be fun?

¹Rawlins (2008) likewise notes that unconditional structures become degraded when combined with interrogative consequents:

- (i) a. #Whether the party is at Alfonso or Joanna’s house, will it last a long time?
b. #Whoever comes to the party, will it last a long time?
c. #No matter who comes to the party, will it last a long time?

(Rawlins 2008: 141)

- b. #Wherever the party is at, will it be fun?

At first glance, this contrast is surprising. If unconditional constructions are interpreted by quantifying over antecedent alternatives, then unconditional questions appear to be compositionally parallel to unconditional statements. One might therefore expect unconditional questions to be as acceptable as their declarative counterparts.

Closer inspection, however, reveals that the degradation is neither absolute nor uniform. Consider the dialogue in (3). Here, the issue introduced by the unconditional antecedent is already salient in the discourse. Although the unconditional question remains somewhat marked, many speakers find it considerably more acceptable than the out-of-the-blue examples in (2).

- (3) A: I think that the party will be at Emma's.
 B: No, didn't they agree that it will be at Luis's?
 A: But Emma's place offers more space!
 C: ?Now, whether or not the party will be at Emma's place, will it be fun?

A similar pattern is observed in examples such as (4), where the antecedent issue appears unrelated to the consequent question.

- (4) Whoever you are, do you need help?

At the same time, not all contexts improve unconditional questions. When the consequent issue depends on the antecedent alternatives, unconditional questions remain strongly degraded.

- (5) Everybody knows that concerts with Ariana Grande are great.
 a. #Whether Ariana Grande is playing or not, do you wanna go to the concert?
 b. #Whoever is playing, do you wanna go to the concert?

The contrast between (3)/(4) and (5) suggests that the acceptability of unconditional questions depends on the relationship between the antecedent issue and the consequent question. The challenge is to explain why some configurations yield relatively acceptable unconditional questions while others remain strongly degraded.

This paper argues that unconditional questions are semantically well formed despite their frequently degraded status. Building on inquisitive semantics and Kaufmann's (2000) stack-based suppositional semantics, I develop a compositional analysis in which unconditional questions are interpreted relative to multiple hypothetical contexts corresponding to antecedent alternatives. I then argue that their

restricted distribution follows from pragmatic constraints on how these hypothetical contexts interact with the consequent question.

More generally, the analysis has implications for the semantics of conditionals. In particular, I argue that both unconditional questions and answer particles in conditional questions provide additional support for a suppositional approach to conditionals.

This paper is structured as follows. Section 2 reviews the empirical and theoretical background and introduces the puzzle posed by the degraded status of unconditional questions. Section 3 develops a compositional semantics for conditional and unconditional questions. Section 4 argues that the restricted distribution of unconditional questions follows from two independent pragmatic factors. Section 5 discusses the implications of the analysis for the semantics of conditionals. Section 6 concludes.

2 Conditional and Unconditional Questions

This section reviews the empirical and theoretical background for the analysis developed in this paper. I first discuss conditional questions and their relevance for suppositional approaches to conditionals. I then review unconditional statements and introduce the puzzle posed by unconditional questions.

2.1 Conditional Questions

Semantic analyses of conditionals are commonly divided into two broad approaches: non-suppositional and suppositional theories. In non-suppositional approaches, conditionals are interpreted as logical implications or universal quantification over antecedent worlds (Lewis 1975, Kratzer 1991). Suppose, for instance, that the antecedent and consequent of the conditional declarative in (6) correspond to propositions p and q , respectively. Under a standard material implication analysis, the sentence is true just in case all p -worlds are also q -worlds, as illustrated in (7).

(6) If the party is at Emma's place, it will be fun.

	p	q	$p \rightarrow q$
	T	T	T
(7)	T	F	F
	F	T	T
	F	F	T

Suppositional approaches to conditionals (Stalnaker 1968, Adams 1965, Mackie 1973, Karttunen 1974, Heim 1982, Gärdenfors 1986: a.o.), by contrast, analyze

- Conditional questions have played an important role in motivating suppositional approaches to conditionals. Consider the conditional interrogative in (9).

- As noted by Isaacs & Rawlins (2008), the interpretation of conditional questions closely follows Ramsey's well-known characterization of conditional reasoning:

- Under a suppositional analysis, the antecedent creates a temporary hypothetical context in which the consequent question is interpreted. This architecture naturally captures the intuition that the question concerns whether q holds *under the assumption that p is true*.

One empirical motivation for this view comes from responses to conditional questions. Consider (11).

- As emphasized by Isaacs & Rawlins (2008), responses such as (11) do not directly answer the consequent question. Rather, they reject the antecedent assumption itself. This observation suggests that the antecedent establishes a temporary inquiry domain distinct from the main discourse context.

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2.2 Unconditional Statements

Unconditional constructions have often been analyzed as conjunctions of conditional statements. Following Rawlins (2008, 2013), an unconditional declarative such as (12) expresses the conjunction of two conditional assertions.

(12) Whether or not the party is at Emma’s place, it will be fun.

Intuitively, (12) corresponds to the conjunction in (13).

- (13) a. If the party is at Emma’s place, it will be fun.
b. If the party is not at Emma’s place, it will be fun.

Because the same consequent proposition holds under both antecedent alternatives, the truth of the consequent is interpreted as independent of how the antecedent issue is resolved.

Importantly, unconditional statements are generally fully acceptable. Unlike unconditional questions, they do not exhibit strong context sensitivity or systematic degradation. This asymmetry between unconditional declaratives and interrogatives constitutes one of the central empirical puzzles addressed in this paper.

2.3 The Puzzle of Unconditional Questions

Suppose unconditional questions have a compositional structure parallel to unconditional statements. Under such a view, the unconditional question in (14) should correspond to a conjunction of two conditional questions.

(14) #Whether or not the party is at Emma’s place, will it be fun?

- (15) a. If the party is at Emma’s place, will it be fun?
b. If the party is not at Emma’s place, will it be fun?

At first glance, this interpretation is perfectly coherent. The construction appears semantically interpretable, and speakers can readily recover the intended meaning. Nevertheless, unconditional questions are frequently judged infelicitous in ordinary out-of-the-blue contexts.

Crucially, however, the degradation is not absolute. As discussed in Section 1, unconditional questions improve in certain discourse environments, including cases where the antecedent issue is independently salient and cases where the consequent issue remains genuinely open across antecedent alternatives, as in (3) and (4). These patterns suggest that the source of degradation cannot be reduced to semantic composition alone.

2.4 From Semantics to Pragmatics

The patterns discussed above raise a fundamental question: are unconditional questions degraded because they are semantically problematic, or because their use is subject to independent pragmatic constraints?

The next section argues that unconditional questions are semantically legitimate by developing a compositional analysis of their interpretation. This result shifts the explanatory burden from semantics to pragmatics.

3 The Semantics of Unconditional Questions

The analysis developed in this paper combines inquisitive semantics (Ciardelli et al. 2015, Ciardelli & Roelofsen 2015, Ciardelli 2016: among others) with Kaufmann’s (2000) stack-based suppositional semantics. Conditional questions are naturally interpreted relative to hypothetical contexts introduced by antecedent clauses, whereas unconditional questions require simultaneous consideration of multiple antecedent alternatives. This section develops a unified framework that captures both phenomena. After introducing the basic machinery of inquisitive semantics and stack-based suppositional semantics, I first show how the framework derives conditional questions and their answer particles. I then extend the analysis to unconditional constructions by introducing multi-stacks.

3.1 Inquisitive Semantics

Inquisitive semantics provides a unified representation of declarative and interrogative meanings. Declaratives contribute information by eliminating incompatible possibilities, while interrogatives introduce issues by distinguishing among alternative possibilities. This dual perspective is particularly useful for the present purposes because conditional and unconditional constructions manipulate hypothetical contexts, whereas questions introduce issues that must remain epistemically unresolved or inquisitive. Combined with the stack-based architecture introduced in Section 3.2, inquisitive semantics provides a natural framework for modeling the interaction between hypothetical reasoning and question interpretation.²

²Isaacs & Rawlins (2008) combine Kaufmann’s stack-based model of conditionals with Groenendijk’s (1999) partition semantics for questions. However, as pointed out by Sano & Hara (2014) and Hara & Sano (2018), this implementation faces technical difficulties. See footnote 8 of Sano & Hara (2014: p. 97) for discussion. Inquisitive semantics provides a more suitable framework for the present purposes because it directly represents questions in terms of possible resolutions and offers a natural way of modeling multiple live epistemic possibilities.

Issues and information states Let W be the set of all possible worlds. As in standard epistemic logic, an information state is identified with a set of possible worlds. Inquisitive semantics extends this perspective by introducing *issues*, which characterize inquisitive epistemic possibilities entertained by discourse participants.

(16) **Definition:** Issue

- a. An information state s is a set of possible worlds, i.e., $s \subseteq W$.
- b. An *issue* $I \subseteq \wp(W)$ is a non-empty downward-closed set of information states.

Intuitively, an issue represents the possible ways in which a question may be resolved. This notion will play a central role in the analysis developed below because the felicity of unconditional questions depends on whether the consequent issue remains inquisitive across antecedent alternatives.

Syntax The language of inquisitive semantics used in this paper is defined as follows.

(17) **Definition:** Syntax

Let P be a set of atomic sentences.

- a. For any $p \in P$, $p \in \mathcal{L}$
- b. If $\phi, \psi \in \mathcal{L}$, $\phi \wedge \psi \in \mathcal{L}$
- c. If $\phi, \psi \in \mathcal{L}$, $\phi \vee \psi \in \mathcal{L}$
- d. If $\phi \in \mathcal{L}$, $\neg\phi \in \mathcal{L}$
- e. If $\phi, \psi \in \mathcal{L}$, $\phi \rightarrow \psi \in \mathcal{L}$

In inquisitive semantics, disjunction is the source of inquisitiveness (Groenendijk 2007, Ciardelli et al. 2019). Polar interrogatives are therefore represented as disjunctions between a proposition and its negation.

(18) **Definition:** Polar interrogatives

$$?\phi := \phi \vee \neg\phi$$

Under this view, asking a question amounts to introducing alternative possibilities whose resolution remains open.

Semantics Unlike standard truth-conditional semantics, inquisitive semantics evaluates both declaratives and interrogatives relative to information states. Declaratives restrict the set of compatible worlds, while interrogatives introduce inquisitive alternatives.

(19) **Definition:** Inquisitive model

An inquisitive model M for a set P of atomic sentences is a tuple $M = \langle W, V \rangle$ such that:

- a. W is a set of possible worlds.
- b. $V : P \rightarrow \wp(W)$ is a valuation function specifying which worlds satisfy each atomic sentence.

The semantic value of a sentence φ , written $\llbracket \varphi \rrbracket$, is an issue: a downward-closed set of information states supporting φ . Declaratives and interrogatives are thus interpreted uniformly as sets of information states.

(20) **Definition:** Semantics

Let M be an inquisitive model and s an information state.

- a. $\llbracket \alpha \rrbracket := \{s \mid w \in V(\alpha) \text{ for all } w \in s\}$
- b. $\llbracket \varphi \wedge \psi \rrbracket := \{s \mid s \in \llbracket \varphi \rrbracket \text{ and } s \in \llbracket \psi \rrbracket\}$
- c. $\llbracket \varphi \vee \psi \rrbracket := \{s \mid s \in \llbracket \varphi \rrbracket \text{ or } s \in \llbracket \psi \rrbracket\}$
- d. $\llbracket \neg \varphi \rrbracket := \{s \mid \forall t \subseteq s [t \neq \emptyset \rightarrow t \notin \llbracket \varphi \rrbracket]\}$
- e. $\llbracket \varphi \rightarrow \psi \rrbracket := \{s \mid \forall t \subseteq s [t \in \llbracket \varphi \rrbracket \rightarrow t \in \llbracket \psi \rrbracket]\}$

Because polar interrogatives are analyzed as disjunctions, their meanings correspond to inquisitive alternative possibilities.

(21) **Definition:** Semantics of polar interrogatives

$$\llbracket ?\varphi \rrbracket := \{s \mid s \in \llbracket \varphi \rrbracket \text{ or } s \in \llbracket \neg \varphi \rrbracket\}$$

A key notion in inquisitive semantics is that of a *possibility*: a maximal information state supporting a sentence.

(22) **Definition:** Possibilities

The maximal states supporting φ are the *possibilities* for φ :

$$\text{possibility}(\varphi) = \{s \mid s \in \llbracket \varphi \rrbracket \text{ and there is no } t \supset s \text{ such that } t \in \llbracket \varphi \rrbracket\}$$

Questions are inquisitive precisely because they introduce multiple possibilities.

(23) **Definition:** Inquisitiveness

- a. φ is inquisitive iff there are at least two possibilities for φ .
- b. φ is non-inquisitive iff there is exactly one possibility for φ .

When φ is inquisitive, its possibilities correspond to the possible resolutions of the issue raised by the question in the sense of Hamblin (1958).

Illustration To illustrate the framework, suppose that $W = \{w_{11}, w_{10}, w_{01}, w_{00}\}$ and that the valuation function satisfies $V(p) = \{w_{11}, w_{10}\}$ and $V(q) = \{w_{11}, w_{01}\}$. Figure 1 illustrates the possibilities associated with several basic expressions. Following standard practice in inquisitive semantics, only maximal possibilities are shown.

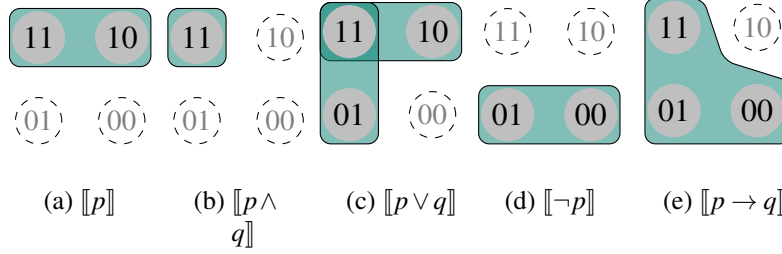


Figure 1 Examples of issues (Following Ciardelli & Roelofsen (2015), only the possibility, i.e., the maximal element of each issue, is shown for readability.)

Among the examples in Figure 1, only $p \vee q$ is inquisitive because it introduces more than one possibility. Declaratives such as p and $p \wedge q$, by contrast, determine a single maximal information state.

Finally, following inquisitive semantics, I treat contexts themselves as issues.

(24) **Definition:** Context

A context $\mathcal{C}$ is a downward-closed set of information states.

A context is inquisitive when it contains more than one maximal possibility.

(25) **Definition:** Inquisitive contexts

- a. $\mathcal{C}$ is inquisitive iff it contains at least two possibilities.
- b. $\mathcal{C}$ is non-inquisitive iff it contains at most one possibility.

Intuitively, an inquisitive context represents an unresolved issue, since multiple possibilities remain available. A non-inquisitive context, by contrast, corresponds to a state in which the issue has already been resolved. This distinction will play an important role later in the paper, where the felicity of unconditional questions is shown to depend on whether the consequent question remains inquisitive across antecedent alternatives.

The representation of contexts introduced here will also become important in the following subsection, where hypothetical updates are modeled using stacks of contexts associated with conditional antecedents.

3.2 Stack-Based Suppositional Semantics

The previous subsection introduced inquisitive semantics as a framework in which both declaratives and interrogatives denote issues. I now combine this framework with a stack-based suppositional semantics in order to model the hypothetical contexts created by conditional and unconditional constructions. As will become clear below, this architecture not only derives the interpretation of conditional questions but also provides a straightforward account of answer particles in hypothetical contexts.

Recall from Section 2.1 that suppositional approaches to conditionals analyze conditionals like (26) dynamically.

(26) If the party is at Emma's place, it will be fun.

Interpreting a conditional involves three conceptual steps as seen in (8), repeated here as (27):

- (27)
- a. A hypothetical context is created by updating the current context with the antecedent ($C \cap P$).
 - b. The hypothetical context is updated with the consequent ($C \cap P \cap Q$).
 - c. The original context incorporates the result of the hypothetical update $((C \cap \bar{P}) \cup (C \cap P \cap Q))$.

To implement these steps formally, I adopt Kaufmann's (2000) stack-based model of hypothetical contexts, originally developed for modal subordination (Landman 1986, Roberts 1996, Frank & Kamp 1997). The central idea is that discourse participants may temporarily reason within derived contexts that differ from the main discourse context. Conditional constructions explicitly introduce such hypothetical contexts.

Updates Recall that a context $\mathcal{C}$ is itself an issue, i.e., a downward-closed set of information states. Declaratives and interrogatives update contexts in the same way: by intersecting the current context with the semantic value of the sentence.

- (28) **Definition:** Update
- $$\mathcal{C} \star \varphi := \mathcal{C} \cap \llbracket \varphi \rrbracket$$

At the initial stage of discourse, the context is maximally uninformative. In other words, discourse participants have not yet committed to any particular information state or issue. This initial context is the trivial issue $\wp(W)$, represented in Figure 2.

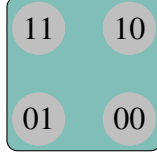


Figure 2 The initial ignorant context

Updating the context with a declarative eliminates worlds incompatible with the asserted proposition. Updating with an interrogative, by contrast, introduces multiple epistemic possibilities corresponding to the possible resolutions of the question.

Stacks and hypothetical contexts To model the hypothetical contexts introduced by conditional antecedents, we extend ordinary contexts to *macro-contexts*, i.e., stacks of contexts.

(29) **Definition:** Macro-context
 $\tau = \langle \mathcal{C}_0, \dots, \mathcal{C}_n \rangle$

As illustrated in Figure 3, each element of the stack is a context, and the topmost context τ_0 is the currently active one.

τ_0	$\mathcal{C}_0$
$\vdots$	$\vdots$
τ_n	$\mathcal{C}_n$

Figure 3 Macro-context τ

To model the interpretation of the *if*-clause in a conditional such as (26), I introduce a push operation, which pushes a temporary context on top of the stack.

(30) **Definition:** push
 For any context $\mathcal{C}$ and macro-context τ :
 $\text{push}(\mathcal{C}, \tau) := \langle \mathcal{C}, \tau \rangle$

The *if*-clause is analyzed as a macro-context change potential (MCCP) that creates a temporary context and restricts it to worlds compatible with the antecedent proposition.

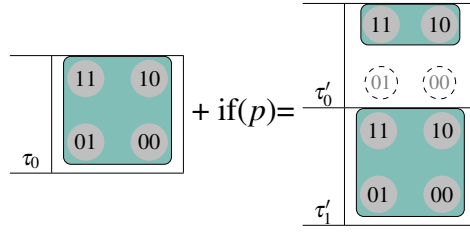
(31) **Definition:** MCCP of an *if*-clauseFor any macro-context τ :

$$\tau + \text{if}(\varphi) := \text{push}(\tau_0 \star \varphi, \tau)$$

Admittance condition:³

$$\tau_0 \star \varphi \neq \emptyset$$

Intuitively, the antecedent temporarily restricts the active context to states that support the antecedent proposition. As illustrated in Figure 4, all states containing worlds incompatible with p —namely w_{01} and w_{00} —are removed from the topmost context.

**Figure 4** $\tau + \text{if}(p)$

The resulting hypothetical context then serves as the domain in which the consequent is interpreted.

Percolation Next, the consequent declarative clause updates the temporary context: $\tau'_0 \star q = \tau'_0 \cap \llbracket q \rrbracket$. Since $w_{10} \notin V(q)$, the update removes all states containing w_{10} .

Crucially, declarative updates are not confined to the temporary context. Rather, information established in the hypothetical context percolates down the stack and constrains the remaining contexts as well.

(32) **Definition:** Percolation

$$\mathcal{C}[\mathcal{C}' \vdash \psi] := \{s \in \mathcal{C} \mid \forall t \subseteq s [t \in \mathcal{C}' \rightarrow t \in \mathcal{C}' \star \psi]\}$$

Intuitively, $\mathcal{C}[\mathcal{C}' \vdash \psi]$ states that discourse participants learn in context $\mathcal{C}$ that the derived context $\mathcal{C}'$ supports ψ .

This operation allows information established within a hypothetical context to constrain the main discourse context while preserving the distinction between the hypothetical context and the actual discourse context.

The corresponding macro-context update is defined as follows.

³The admittance condition encodes the presupposition that the propositional content of the antecedent is possible, an assumption that has often been adopted since [Stalnaker \(1968\)](#).

(33) **Definition:** MCCC of UPDATE
 $\tau + \text{UPDATE}(\varphi) := \langle \tau_i[\tau_0 \vdash \varphi] \rangle_{0 \leq i < n}$

Unlike Isaacs & Rawlins (2008), who distinguish assertive and inquisitive updates at the level of macro-contexts, the present framework treats both declarative and interrogative updates uniformly. In both cases, updating amounts to intersecting the current context with the semantic value of the sentence.⁴

The effect of MCCC of UPDATE is illustrated in Figure 5.

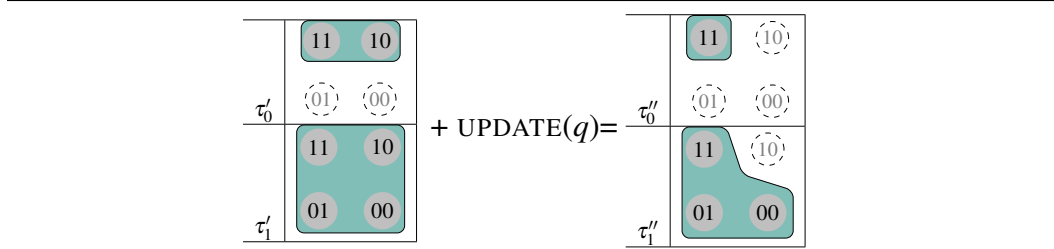


Figure 5 Update of the consequent declarative

The consequent proposition is first established within the hypothetical context and then percolates to the lower context, yielding the update in (34).

(34) $\tau + \text{if}(p) + \text{UPDATE}(q) = \langle \tau_i[\tau_0 \star p \vdash q] \rangle_{0 \leq i \leq n}$

After the consequent has been evaluated, the temporary hypothetical context is normally discarded unless it is required for subsequent discourse, as in modal subordination. Following Isaacs & Rawlins (2008), I assume a POP operation.⁵ The result is equivalent to updating the original context with the ordinary, i.e., non-suppositional conditional ‘ $\rightarrow$ ’:

(35) $\mathcal{C}[\mathcal{C} \star p \vdash q] = \mathcal{C} \star (p \rightarrow q).$

The stack-based derivation therefore reproduces the familiar interpretation of conditionals while preserving the distinction between actual and hypothetical contexts throughout the derivation. This distinction will become particularly important in the analysis of conditional questions, where interrogative updates apply within hypothetical contexts and give rise to characteristic patterns of answer interpretation.

⁴The update rule adopted here is a reformulation of Isaacs and Rawlins’ (2008) notion of *support* and can be viewed as a natural extension of Kaufmann’s (2000) Conclude operation. As shown in Appendix A.1, the operation derives the expected conditional interpretation: $\mathcal{C}[\mathcal{C} \star p \vdash q] = \mathcal{C} \star (p \rightarrow q)$. See also Sano & Hara (2014) for discussion of a technical problem with the original formulation of support.

⁵See Appendix A.2 for the definition of POP and an illustration.

3.3 Conditional Questions and Answer Particles

Before turning to unconditional questions, let us first consider ordinary conditional questions such as (36).

(36) If the party is at Emma's place, will it be fun?

Conditional questions illustrate how interrogative updates interact with hypothetical contexts in the stack-based framework developed above. In particular, they raise an important question concerning the interpretation of answer particles. Intuitively, responses such as *yes* and *no* retain their ordinary meanings while receiving the expected interpretations in conditional environments. In this section, I show how the stack-based framework derives this result compositionally. The analysis will then serve as the basis for the extension to unconditional questions developed in the following section.

3.3.1 Deriving Conditional Questions

The antecedent first creates a hypothetical context restricted to worlds in which the party is at Emma's place. The consequent interrogative is then interpreted relative to this derived context. Since the interrogative introduces multiple possibilities, the topmost context becomes inquisitive. Crucially, inquisitive updates may also percolate down the stack. The derivation of a conditional question proceeds as follows:

$$(37) \quad \tau + \text{if}(p) + \text{UPDATE}(?q) = \langle \tau_i[\tau_0 \star p \vdash ?q] \rangle_{0 \leq i \leq n}$$

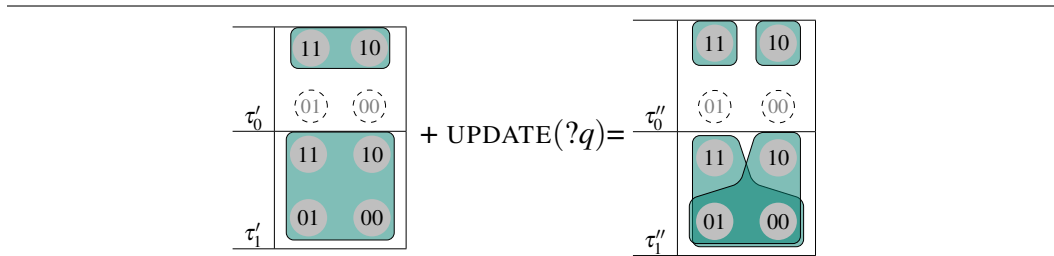


Figure 6 $\tau + \text{if}(p) + \text{UPDATE}(?q)$

Suppose that q is true in w_{11} but false in w_{10} . Since these two worlds resolve the question $?q$ differently, the state $\{w_{11}, w_{10}\}$ does not support the interrogative: $\{w_{11}, w_{10}\} \notin \llbracket ?q \rrbracket$. Consequently, $\{w_{11}, w_{10}\} \notin \tau_0 \star p \star ?q$, and the state containing both worlds is removed from the temporary context. As shown in Figure 6, the

hypothetical context is thereby split into two maximal possibilities, corresponding to the positive and negative resolutions of the question.

The effect of the interrogative update also percolates down the stack, yielding a lower context with overlapping possibilities:

$$(38) \quad \mathcal{C}[\mathcal{C} \star p \vdash ?q] = \{s \in \mathcal{C} \mid \forall t \subseteq s [t \in \mathcal{C} \star p \rightarrow t \in \mathcal{C} \star p \star ?q]\}.$$

3.3.2 The Interpretation of Answer Particles

After the interrogative update, the topmost context remains inquisitive and awaits resolution. Following standard assumptions, I treat *yes* and *no* as updates with q and $\neg q$, respectively. A positive answer therefore resolves the issue by eliminating the possibility incompatible with q .

$$(39) \quad \tau + \text{if}(p) + \text{UPDATE}(?q) + \text{UPDATE}(q) = \langle \tau_i[\tau_0 \star p \star ?q \vdash q] \rangle_{0 \leq i \leq n}$$

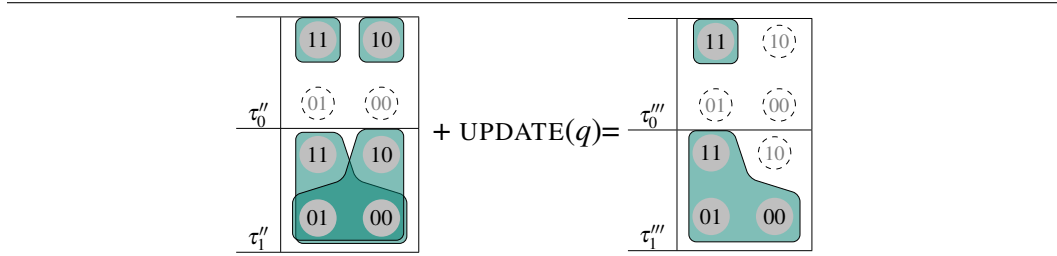


Figure 7 A *yes* answer given to the conditional question

Updating the topmost context with q removes the possibility containing w_{10} , thereby resolving the issue introduced by the interrogative. This effect percolates to the lower context as well, yielding the macro-context shown in Figure 7.

An advantage of the present account is that answer particles retain their ordinary meanings. Unlike the analysis of [Velissaratou \(2000\)](#), no special meanings such as $p \rightarrow q$ or $p \rightarrow \neg q$ need to be assigned to *yes* and *no*.⁶

⁶After the issue has been resolved, the temporary context may be removed by the POP operation. Isaacs and Rawlins (2008), however, argue that derived contexts should not be popped while the topmost context remains inquisitive:

- (i) INQUISITIVE CONSTRAINT
A macro-context may not be popped if the top element is inquisitive. ([Isaacs & Rawlins 2008](#): (49), p. 294)

Since the present paper does not rely on the subsequent manipulation of inquisitive derived contexts, I leave the broader question of when inquisitive contexts should be retained for future research.

3.3.3 Why Suppositional Semantics Matters

The preceding discussion showed that answer particles in conditional questions can retain their ordinary meanings while receiving the expected interpretations in conditional environments. This result already points toward a suppositional treatment of conditionals.⁷

Additional evidence comes from the behavior of responses to conditional questions. Isaacs and Rawlins (2008) observe that the possibility of denial of the antecedent as in (40) poses a challenge for non-suppositional approaches.

- (40) a. If the party is at Emma's place, will it be fun?
 b. The party is not at Emma's place.

Intuitively, the response in (40-b) does not answer the question. Rather, it rejects the presupposition that the antecedent remains a live possibility. This observation is difficult to capture in approaches that treat the denial of the antecedent as one of the semantic alternatives associated with the question.

The analysis thus provides additional support for a suppositional treatment of conditionals.⁸ On this view, the apparent conditional interpretation of answer particles arises from the temporary hypothetical context introduced by the antecedent rather than from any special lexical meaning associated with *yes* and *no*. As we will see in the next section, the same architecture also provides the foundation for a compositional analysis of unconditional questions.

3.4 Deriving Unconditional Questions

The previous section showed that conditional questions can be interpreted by evaluating an interrogative consequent relative to a temporary hypothetical context introduced by the antecedent. I now extend this architecture to unconditional constructions. Building on Rawlins' (2008, 2013) insight that unconditional constructions involve quantification over antecedent alternatives, I propose that unconditionalization creates multiple hypothetical contexts simultaneously. Whereas ordinary

⁷The need for hypothetical contexts that persist inter-sententially is independently recognized for modal subordination (Landman 1986, Roberts 1996, Frank & Kamp 1997, Kaufmann 2000). In (i), (i-b) affects the hypothetical context created by the antecedent of (i-a), not the utterance context, which includes worlds where Edna does not forget to fill the birdfeeder.

- (i) a. If Edna forgets to fill the birdfeeder, she will feel very bad.
 b. The birds will get hungry. (Roberts 1996: 683)

⁸See Section 5 for further discussion.

conditionals require a single temporary context, unconditional constructions require a collection of hypothetical contexts corresponding to the antecedent alternatives.

As in the case of ordinary conditional constructions, inquisitive semantics provides a uniform update procedure for both declarative and interrogative consequents. The main task is therefore to represent multiple hypothetical contexts and to ensure that updates apply to all of them in a systematic way. To accomplish this, I introduce the notion of a *multi-stack*, a collection of stacks that simultaneously tracks the interpretation of each antecedent alternative.

3.4.1 Multi-Stacks for Unconditionals

Recall that ordinary conditional constructions create a single hypothetical context on top of a stack. Unconditional constructions differ in that they introduce multiple antecedent alternatives. To represent the hypothetical contexts associated with these alternatives, I introduce the notion of a *multi-stack*, that is, a sequence of stacks.

- (41) **Definition:** Multi-stack
 $\mathcal{T} = \langle \tau^{(0)}, \tau^{(1)}, \dots, \tau^{(n-1)} \rangle$
 where each $\tau^{(i)}$ is a macro-context.

As a first step, the initial stack is copied by an *n*-COPY operation.

- (42) **Definition:** *n*-COPY
 For any macro-context τ :
 $\tau + n\text{-COPY} := \langle \tau^{(0)}, \dots, \tau^{(n-1)} \rangle$
 where $\tau = \tau^{(0)} = \dots = \tau^{(n-1)}$

The unconditional antecedent then distributes the antecedent alternatives across the copied stacks.

- (43) **Definition:** MCCP of unconditional antecedents
 For each $\alpha_i \in \text{Alt}_{\text{ant}}(\varphi)$:
 $\tau + \text{unconditional}(\varphi) := \langle \tau^{(0)} + \text{if}(\alpha_0), \dots, \tau^{(n-1)} + \text{if}(\alpha_{n-1}) \rangle$,
 where $\text{Alt}_{\text{ant}}(\varphi)$ denotes the set of antecedent alternatives introduced by unconditionalization.

For polar antecedents such as *whether or not p*, two alternatives are introduced: $\text{Alt}_{\text{ant}}(p) = \{p, \neg p\}$. Thus, unconditional questions create two hypothetical contexts simultaneously: one in which *p* holds and another in which $\neg p$ holds.

If the unconditional antecedent is a polar question, *whether-or-not-p*, $n = 2$, thus (43) reduces to (44):

- (44) **Definition:** MCCP of an *whether-or-not* φ
 For a macro-context τ and an unconditional antecedent (whether or not φ):
 $\tau + \text{whether-or-not}(\varphi) := \langle \tau^{(0)} + \text{if}(\varphi), \tau^{(1)} + \text{if}(\neg\varphi) \rangle$,
 where $\langle \tau^{(0)}, \tau^{(1)} \rangle = \tau + 2\text{-copy}$.

3.4.2 Unconditional Statements

Let us consider the unconditional antecedent of (45) with the same model as before:
 $V(p) = \{w_{11}, w_{10}\}$, and $V(q) = \{w_{11}, w_{01}\}$, where p = ‘The party is at Emma’s place’ and q = ‘It will be fun’.

- (45) Whether or not the party is at Emma’s place, it will be fun.

The initial context has the form of a single stack. The processing of the *whether-or-not* adjunct creates two stacks, corresponding to the alternatives p and $\neg p$. In other words, a *whether-or-not*-adjunct denotes a macro-context change potential which creates a multi-stack and performs an update over the created multi-stack. Thus, the main macro-context is first rendered into a sequence of macro-contexts and (44) performs the MCCP of an *if*-clause (31) in each of the macro-contexts. The definitions of an *if*-clause (31) and the push operator (30) are directly carried over to the current framework. In effect, one temporary context is created in each stack, the top member $\tau_1'^{(0)}$ of $\tau'^{(0)}$ is updated with p , and $\tau_1'^{(1)}$ is updated with $\neg p$, as depicted in Figure 8.

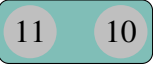
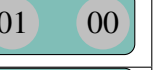
	$\tau'^{(0)}$	$\tau'^{(1)}$
		
$\tau'_0 :$		
$\tau'_1 :$		

Figure 8 $\tau + \text{whether-or-not}(p)$

Once the antecedent alternatives have created a multi-stack, the consequent can be interpreted in exactly the same way as in ordinary conditionals. To implement this idea, I define an operation MSUPDATE. Note that, like UPDATE in (33), MSUPDATE

is defined uniformly for both declarative and interrogative updates by applying the corresponding update operation to each stack in the multi-stack:

(46) **Definition:** MSUPDATE

For a multi-stack $\mathcal{T}$:

$$\mathcal{T} + \text{MSUPDATE}(\varphi) := \langle \tau^{(0)} + \text{UPDATE}(\varphi), \dots, \tau^{(n)} + \text{UPDATE}(\varphi) \rangle$$

The information percolates to the lower contexts through the MCCP of UPDATE and the resulting multi-stack is shown in Figure 9.

	$\tau''^{(0)}$	$\tau''^{(1)}$
$\tau_0'' :$	11 10	11 10
	01 00	01 00
$\tau_1'' :$	11 10 01 00	11 10 01 00

Figure 9 $\tau + \text{whether-or-not}(p) + \text{MSUPDATE}(q)$

Since the temporary hypothetical contexts are no longer needed, they may be removed by a multi-stack version of the POP operation and the resulting macro-contexts merged into a single context.⁹

Consequently, the final context is identical to the one obtained by updating the initial context with the simple declarative sentence in (47).

(47) The party will be fun.

The analysis therefore derives the consequent entailment of unconditional statements. In this respect, the present suppositional account captures the same effect as Rawlins' (2008, 2013) non-suppositional analysis, where unconditionality arises from universal quantification over alternative conditionals.

This result establishes the semantic treatment of unconditional statements. The next subsection shows that the same multi-stack architecture extends naturally to unconditional questions, where interrogative updates apply simultaneously across antecedent alternatives.

⁹See Appendix A.3 for the definitions of MSpop and merge and the derivation of consequent entailment.

3.4.3 Unconditional Questions

The previous subsection showed how unconditional declaratives can be interpreted within a multi-stack architecture. I now extend the same machinery to unconditional questions such as (48). The analysis shows that unconditional questions are compositionally interpretable within the present framework. Their restricted distribution is therefore attributed to pragmatic rather than semantic factors, discussed in Section 4.

(48) #Whether or not the party is at Emma's place, will it be fun?

The *whether-or-not*-clause creates a sequence of stacks in the same fashion as depicted in Figure 8 for the assertion case. The MSUPDATE of the consequent defined in (46) then operates over those contexts. Figure 10 illustrates the creation of temporary contexts and how the entire multi-stack is updated.

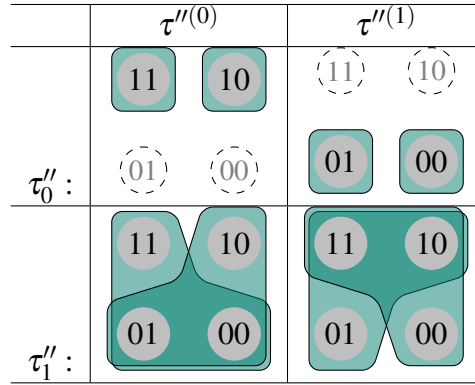


Figure 10 $\tau + \text{whether-or-not}(p) + \text{MSUPDATE}(?q)$

Unlike unconditional declaratives, unconditional questions leave each hypothetical context inquisitive. The consequent interrogative partitions the possibilities within both hypothetical contexts, creating unresolved issues corresponding to the possible answers to the question. The resulting multi-stack therefore represents two simultaneously maintained hypothetical contexts, each associated with a distinct antecedent alternative and containing the issue introduced by the consequent question. This corresponds to the paraphrases in (49).

- (49) a. If the party is at Emma's place, will it be fun?
 b. If the party is not at Emma's place, will it be fun?

This derivation shows that unconditional questions are compositionally interpretable within the present framework. The semantics therefore predicts that they

are semantically legitimate. The remaining question is why such constructions are often judged degraded in actual discourse. I turn to this issue in the next section.

4 Two Sources of Pragmatic Degradation

The preceding sections showed that unconditional questions are compositionally interpretable within a framework that combines inquisitive semantics with stack-based suppositional semantics. Their meanings can be derived in a fully systematic way, suggesting that their often degraded status cannot be attributed to semantic ill-formedness.

I argue instead that the degradation is pragmatic in nature and arises from two distinct sources. When the consequent issue depends on the antecedent issue, unconditionalization may fail to sustain an inquisitive issue across antecedent alternatives, giving rise to a violation of an Extended Epistemic Bridge Requirement. When the antecedent issue and the consequent issue are independent, unconditionalization contributes no additional semantic content beyond that of the corresponding ordinary question. In such cases, the use of the more complex unconditional form requires independent pragmatic motivation, which I argue is provided by relational indifference.

4.1 Dependent Contexts and the Extended Epistemic Bridge

I begin with cases in which the consequent issue depends on the antecedent issue. Such cases are particularly revealing because they illustrate a type of degradation that cannot be reduced to semantic ill-formedness.

Consider again (5-a), repeated here as (50).

- (50) Everybody knows that concerts with Ariana Grande are great.
 #Whether Ariana Grande is playing or not, do you want to go to the concert?

As shown in the previous sections, the semantics assigns a well-defined interpretation to (50). The unconditional antecedent introduces multiple hypothetical contexts, and the consequent question is evaluated relative to each of them. Nothing in the compositional derivation predicts that the question should be anomalous.

Nevertheless, speakers often find examples such as (50) degraded. Intuitively, the problem is that the antecedent alternatives do not contribute equally to the inquiry raised by the consequent question. Given the background assumption that concerts featuring Ariana Grande are enjoyable, the answer to the consequent question is no longer genuinely open in all antecedent alternatives. The felicity of the question therefore appears to depend not only on its semantic interpretation but also on the epistemic status of the alternatives introduced by unconditionalization.

The following subsections develop this intuition in more detail. Building on Theiler’s (2021) Epistemic Bridge approach to interrogatives, I argue that unconditional questions require all antecedent alternatives to sustain a genuinely unresolved, i.e., inquisitive, issue with respect to the consequent question. When this requirement is not met, the question becomes pragmatically degraded.

4.1.1 Theiler’s Epistemic Bridge

A useful starting point is Theiler’s (2021) Epistemic Bridge account of interrogative felicity. Theiler’s proposal starts from the observation that asking a question presupposes a particular relationship between the speaker’s epistemic state and the alternatives introduced by the question.

One phenomenon that motivates Theiler’s proposal is the universal projection of presuppositions in questions. To illustrate, it is well known that in (51), the presupposition introduced by the factive verb *regret* projects universally, so that the question presupposes that Mary believes that Bill invited all ten people:

- (51) Who among these ten people does Mary regret that Bill invited?
 $\rightsquigarrow$ Mary believes that Bill invited all ten of these people. (Abrusán 2014)

According to Theiler (2021), this universal projection effect follows from a general condition on question felicity. A rational questioner seeks to avoid situations in which the common ground cannot be coherently updated with the true answer. Consequently, the context set must already entail the presuppositions of all answers that the speaker considers possibly true.

Theiler’s Epistemic Bridge requirement is defined as in (52).¹⁰

- (52) **Epistemic Bridge**
 A question Q is felicitous relative to a context set c only if the presuppositions of all answers that the speaker considers possibly true are entailed by the context set (i.e., only if $\forall p \in Q \mid \sigma : c \subseteq p\pi$).
 (Theiler 2021: 262)

Informally, the requirement ensures that every answer under consideration is licensed by the discourse context. Questions are felicitous only when the common ground is compatible with all answers that remain epistemically possible for the speaker. In the case of (51), every answer that remains epistemically possible for

¹⁰The set $Q \mid \sigma$ excludes those answers whose at-issue content is inconsistent with the speaker’s information state σ :

- (i) $Q \mid \sigma := , p \in Q \mid p\alpha \cap \sigma \neq \emptyset,$ (Theiler 2021: 261)

the speaker carries a presupposition of the form “Mary believes that Bill invited x .” The Epistemic Bridge therefore requires the context set to support the presuppositions associated with all such answers, yielding the universal projection effect. The intuition behind the Epistemic Bridge can be extended to unconditional questions.

4.1.2 The Extended Epistemic Bridge Requirement

Theiler’s Epistemic Bridge is formulated for ordinary questions.¹¹ In such cases, the relevant epistemic alternatives are the possible answers to the question. Unconditional questions, however, introduce an additional layer of structure. Besides the answer alternatives introduced by the consequent question, unconditionalization creates multiple hypothetical contexts corresponding to distinct antecedent alternatives.

This difference suggests that the felicity of unconditional questions depends not only on which answers remain epistemically possible, but also on how the consequent question behaves across antecedent alternatives. In particular, the consequent question must remain inquisitive (see (23) and (25)) in each hypothetical context introduced by unconditionalization. If one antecedent alternative already determines the answer to the consequent question, updating that context with $?q$ yields a non-inquisitive context. Consequently, unconditionalization fails to establish an inquisitive issue across all antecedent alternatives.

I therefore propose the following extension of Theiler’s Epistemic Bridge requirement.

(53) **Extended Epistemic Bridge Requirement (EEBR)**

$\forall a \in \text{Alt}_{\text{ant}}(p) : (\mathcal{C} \star a) + ?q$ is inquisitive,
where $\text{Alt}_{\text{ant}}(p)$ denotes the antecedent alternatives introduced by unconditionalization.

Informally, the consequent question must remain unresolved in every relevant antecedent alternative. That is, updating a hypothetical context with $?q$ must yield at least two possibilities, corresponding to distinct potential resolutions of the issue. When one antecedent alternative already determines the answer to the consequent question, the interrogative update fails to create an inquisitive context, the EEBR is violated, and the resulting unconditional question becomes pragmatically degraded.

For ordinary polar questions, the EEBR collapses to the intuition underlying Theiler’s Epistemic Bridge: the question must introduce an unresolved issue. Un-

¹¹More precisely, Theiler (2021) develops the Epistemic Bridge account to explain why ordinary (canonical) questions exhibit universal presupposition projection, whereas non-canonical questions such as quiz questions and rhetorical questions do not.

conditional questions, however, require this condition to hold separately in each antecedent alternative introduced by unconditionalization. Consequently, the EEBR imposes a stronger condition: the consequent question must remain inquisitive in every antecedent alternative.

The EEBR predicts that unconditional questions become degraded when one or more antecedent alternatives fail to sustain an inquisitive context with respect to the consequent question. Consider the Areana Grande example (50). Here, the discourse context already supports the conditional proposition, $p \rightarrow q$, where p denotes the proposition that Ariana Grande is playing and q denotes the proposition that the concert will be fun. As a result, whenever p is true, q is already established.

Figure 11 represents a context that supports $p \rightarrow q$. Since the world w_{10} is excluded, every p -world in the context is also a q -world.

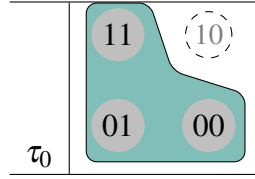


Figure 11 The pre-update context τ supporting $p \rightarrow q$

The unconditional antecedent then creates two hypothetical contexts corresponding to the p - and $\neg p$ -alternatives. The subsequent interrogative update with $?q$ yields the multi-stack shown in Figure 12.

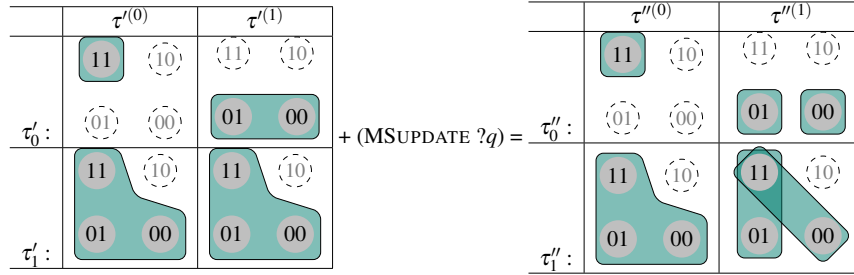


Figure 12 $\tau + \text{whether-or-not}(p) + \text{MSUPDATE}(?q)$

Crucially, the p -alternative already contains only q -worlds before the interrogative update applies. As a result, $(\mathcal{C} \star p) \models q$, and updating with $?q$ fails to create an inquisitive context: $(\mathcal{C} \star p) + ?q$ is non-inquisitive. Since the Extended Epistemic Bridge Requirement requires the consequent question to remain inquisitive in every antecedent alternative, the p -alternative violates the requirement. The unconditional

question therefore becomes pragmatically degraded despite being compositionally well formed.

4.2 Independent Contexts and Relational Indifference

The previous subsection showed that EEER violations can account for some degraded unconditional questions. However, they cannot account for all cases. I argue that the remaining cases involve relational indifference (Rawlins 2008, 2013), a Gricean inference that the antecedent issue is irrelevant to resolving the consequent question.

4.2.1 Semantic Equivalence to Ordinary Questions

Consider again the out-of-the-blue example in (14), repeated here as (54).

(54) #Whether or not the party is at Emma's place, will it be fun?

Unlike the Ariana Grande example, neither antecedent alternative settles the answer to the consequent question. The EEER is therefore satisfied. Nevertheless, the unconditional question remains degraded. This shows that satisfying the EEER is not sufficient for unconditional questions to be felicitous. A second pragmatic factor is therefore needed.

The source of degradation cannot lie in a failure of the consequent question to remain inquisitive. Rather, when the antecedent issue $?p$ and the consequent issue $?q$ are independent, unconditionalization does not affect the denotation of the consequent question. The antecedent alternatives merely create multiple hypothetical contexts in which the same issue remains inquisitive. In other words, an unconditional question of the form *whether or not p*, $?q$ is equivalent to $?q$. This equivalence is stated in Theorem (55).¹²

(55) Theorem

Let p and q be independent in $\mathcal{C}$, and let p be consistent with $\mathcal{C}$.
Then, $\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q)$ is equivalent to $\mathcal{C} \Rightarrow ?q$.

The unconditional question and the corresponding ordinary question therefore raise the same issue and receive the same semantic interpretation. The degradation of out-of-the-blue unconditional questions cannot therefore be attributed to their semantic content. If speakers nevertheless choose the more complex unconditional form, that choice must serve some independent communicative purpose. The next subsection argues that this purpose is captured by Rawlins' (2008, 2013) notion of relational indifference.

¹²See Appendix B for a formal proof.

4.2.2 Relational Indifference as a Gricean Inference

The semantic equivalence established above raises an obvious question. If an unconditional question and its corresponding ordinary question introduce the same issue, why would a speaker choose the more complex unconditional form?

A natural answer follows from standard Gricean reasoning about markedness. When two expressions are semantically equivalent, a speaker's choice of the more marked expression is normally taken to convey additional information.¹³ Otherwise, the use of the marked form would be pragmatically unmotivated.

This reasoning is closely related to Rawlins' (2008, 2013) notion of *relational indifference*. In the case of unconditional declaratives, unconditionalization signals that the truth of the consequent does not depend on which antecedent alternative holds. To illustrate, Rawlins argues that an utterance of (56) conveys that "the resolution of the issue described by the adjunct ... is independent of the truth of the main clause" (Rawlins 2013: 113). Accordingly, (56) gives rise to the inference indicated after $\rightsquigarrow$.

- (56) Whether Alfonso is great at his job or not, we should fire him.
 $\rightsquigarrow$ It does not matter whether he is great at his job for the purposes of this decision. (Rawlins 2013: 113)

I propose that a similar inference arises in unconditional questions. When an unconditional question is used despite being semantically equivalent to the corresponding ordinary question, the hearer searches for a reason why the antecedent issue has been explicitly introduced. The resulting inference is that the antecedent issue is not relevant to resolving the consequent question. By mentioning the antecedent alternatives and then asking the consequent question, the speaker signals that the answer should be sought independently of how the antecedent issue is resolved.

Under this view, relational indifference is not part of the semantic content of unconditional questions. Rather, it is a pragmatic inference arising from the speaker's choice of a marked form where a simpler alternative would have expressed the same issue. The unconditional construction therefore communicates not only the consequent question itself, but also the speaker's stance that the antecedent issue should be set aside for the purposes of resolving that question.

¹³This idea is formulated more explicitly by Horn (1984) as in (i):

- (i) Horn's Division of Pragmatic Labor The use of a marked expression when a corresponding unmarked alternative expression is available tends to be interpreted as conveying a marked message, one that the unmarked alternative would not or could not have conveyed.

4.2.3 Why Salient Issues Matter

The analysis developed above predicts that relational indifference can arise only when the antecedent issue is already salient in the discourse. If unconditionalization signals that the antecedent issue should be set aside for the purposes of resolving the consequent question, there must first be an antecedent issue available for the hearer to identify.

This prediction is illustrated most clearly by examples such as (4), repeated here as (57).

- (57) Whoever you are, do you need help?
 ↪ It does not matter who you are, for determining whether help is needed.

Here, the identity of the addressee is explicitly raised by the unconditional antecedent, allowing the speaker to convey that this issue is irrelevant to determining whether help is needed. The resulting inference of relational indifference is therefore straightforward to recover.

A similar effect can be observed in (3), repeated here as (58). The interlocutors are already engaged in a disagreement about the location of the party. The unconditional question can therefore be interpreted as signaling that, regardless of how that issue is resolved, the remaining question concerns whether the party will be fun.

- (58) A: I think that the party will be at Emma's.
 B: No, didn't they agree that it will be at Luis's?
 A: But Emma's place offers more space!
 C: ?Now, whether or not the party will be at Emma's place, will it be fun?
 ↪ It does not matter where the party is, for resolving whether the party will be fun.

In both cases, the antecedent issue is already available in the discourse, and the unconditional construction serves to indicate that it should be disregarded in resolving the consequent question. The resulting inference of relational indifference is therefore easy for the hearer to recover.

By contrast, in out-of-the-blue examples such as (54), the antecedent issue is not independently salient. The hearer is therefore left without a clear explanation for why the antecedent alternatives have been introduced. Since the corresponding ordinary question would express the same issue more directly, the unconditional form appears pragmatically unmotivated and is consequently judged degraded.

5 Consequences for the Semantics of Conditionals

The analysis developed in this paper has implications beyond unconditional questions. I argue that both the Extended Epistemic Bridge Requirement and the interpretation of answer particles provide additional support for a suppositional approach to conditionals, according to which antecedents introduce temporary hypothetical contexts for evaluating consequent information and inquiry. The remainder of this section explores these implications.

5.1 Multiple Hypothetical Contexts

One consequence of the preceding discussion is that the felicity of unconditional questions cannot be evaluated solely with respect to a single global context. The Extended Epistemic Bridge Requirement (53) must be checked separately for each antecedent alternative introduced by unconditionalization.

Recall the Ariana Grande example (50) discussed in Section 4.1. The unconditional question is degraded because the antecedent alternative in which Ariana Grande is playing already settles the consequent issue, whereas the alternative in which she is not playing does not. Detecting this asymmetry requires evaluating the antecedent alternatives independently.

More generally, the EEBR quantifies over antecedent alternatives and asks whether the consequent question remains inquisitive in each of them: $\forall a \in \text{Alt}_{\text{ant}}(p) : (\mathcal{C} \star a) + ?q$ is inquisitive. That is, each antecedent alternative must support at least two possibilities after updating with the consequent question.

The evaluation of this condition therefore presupposes a representation in which antecedent alternatives are maintained as distinct hypothetical contexts. The multi-stack architecture introduced in (41) provides exactly such a representation: each antecedent alternative is associated with its own hypothetical context, allowing the EEBR to be evaluated locally. The analysis of unconditional questions thus provides independent motivation for representing antecedent alternatives as distinct hypothetical contexts rather than collapsing them into a single global representation.¹⁴

¹⁴The notion of a multi-stack may also prove useful for analyzing modal splitting, a form of modal subordination discussed by Landman (1986). As illustrated in (i), the interpretation of subsequent discourse may require simultaneous access to multiple hypothetical contexts:

- (i) a. You will stay unmarried or you will marry a trump. You'll become a nun, or the trump will beat you regularly. Either way, you'll have a miserable life.
- b. Call this number. The phone will be answered by either a doctor or a [nurse]. The doctor can tell you right away what's the matter with you, or the nurse can make an appointment for you. (Landman 1986: 205)

I argue in the next subsection that stack-based suppositional semantics provides a particularly natural implementation of this idea.

5.2 Suppositional Semantics Revisited

The preceding subsection argued that the analysis of unconditional questions requires access to individual antecedent alternatives and their associated hypothetical contexts. This conclusion does not by itself determine which theory of conditionals is correct. Nevertheless, the present analysis provides additional support for a suppositional approach to conditionals.

Under suppositional semantics, antecedents introduce temporary hypothetical contexts that serve as local domains for interpretation. This architecture makes it possible to evaluate both declarative and interrogative consequents relative to antecedent-specific information states. In the case of unconditional constructions, the same idea naturally extends to multiple antecedent alternatives, each of which gives rise to its own hypothetical context.

The resulting framework yields a unified treatment of several phenomena discussed in this paper. First, as shown in Section 3.3.1, conditional questions can be interpreted by evaluating the consequent issue relative to a hypothetical context introduced by the antecedent. Second, as argued by Isaacs & Rawlins (2008) and illustrated in Section 3.3.2, answer particles like *yes* and *no* retain their ordinary meanings because they update a local hypothetical context rather than encode conditional meanings themselves. Finally, as shown in Section 3.4.3, unconditional questions can be analyzed in terms of multiple hypothetical contexts, with the consequent issue evaluated relative to each of them.

These results contrast with approaches that do not make hypothetical contexts available as explicit objects of interpretation. In such approaches, the connection between antecedent alternatives and local inquiry states is considerably less transparent. By representing hypothetical contexts directly, suppositional semantics provides a natural foundation for both the semantic derivation of unconditional questions and the pragmatic constraints governing their use.

The argument developed here is therefore not merely that unconditional questions can be accommodated within a suppositional framework. Rather, the interaction of conditional questions, answer particles, and unconditional questions suggests that hypothetical contexts play an active role in interpretation. Suppositional semantics captures this role directly and thus offers a particularly attractive framework for understanding the semantics and pragmatics of conditional constructions.

6 Conclusion

This paper has argued that unconditional questions are semantically well formed despite their frequently degraded status. Building on inquisitive semantics and Kaufmann’s (2000) stack-based suppositional semantics, I developed a compositional account of conditional questions, unconditional statements, and unconditional questions. In particular, I proposed a multi-stack architecture in which antecedent alternatives create distinct hypothetical contexts that are updated simultaneously by the consequent.

The analysis showed that unconditional questions do not require any special semantic restrictions. Like unconditional statements, they can be compositionally interpreted by distributing antecedent alternatives across multiple hypothetical contexts and evaluating the consequent relative to each of them. The resulting derivation predicts that unconditional questions are semantically legitimate expressions.

The central puzzle is therefore pragmatic rather than semantic. I argued that two independent factors contribute to the degraded status of unconditional questions. First, when the consequent issue depends on the antecedent alternatives, one or more alternatives may fail to sustain a genuinely inquisitive issue. To capture this restriction, I proposed the Extended Epistemic Bridge Requirement, which extends Theiler’s (2021) Epistemic Bridge account to unconditional questions. Second, when the antecedent and consequent issues are independent, unconditional questions are semantically equivalent to their corresponding ordinary questions. Their use therefore invites a Gricean inference that the antecedent issue is irrelevant to resolving the consequent issue, giving rise to a form of relational indifference in the sense of Rawlins (2008, 2013). This account further explains why unconditional questions improve when the antecedent issue is already salient in the discourse.

Beyond unconditional questions, the analysis has broader implications for the semantics of conditionals. The Extended Epistemic Bridge Requirement must be evaluated relative to individual antecedent alternatives, motivating a representation in which alternative antecedents correspond to distinct hypothetical contexts. Together with the analysis of answer particles in conditional questions, the present results provide additional support for a suppositional approach to conditionals, in which hypothetical contexts play an active role in interpretation.

More generally, the paper suggests that the felicity of interrogatives depends not only on the issues they introduce, but also on how those issues are distributed across alternative contexts. Understanding this interaction between inquiry, hypothetical reasoning, and discourse structure remains an important task for future research.

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A Formal illustration of derivations of (un)conditional statements/questions

A.1 Support and Conditional Updates

This appendix illustrates how the notion of support used in the present paper derives ordinary conditional interpretations.

The operation $\mathcal{C}[\mathcal{C}' \vdash \psi]$ can be read informally as follows:

- (59) Discourse participants learn in context $\mathcal{C}$ that the derived context $\mathcal{C}'$ supports ψ .

The definition is a reformulation of Isaacs and Rawlins' (2008) notion of support and a natural extension of Kaufmann's (2000) Conclude operation.

$$(60) \quad \mathcal{C}[\mathcal{C} \star p \vdash q] = \mathcal{C} \star (p \rightarrow q)$$

To see this, suppose that $\mathcal{C}$ is the trivial issue shown in Figure 13(a).

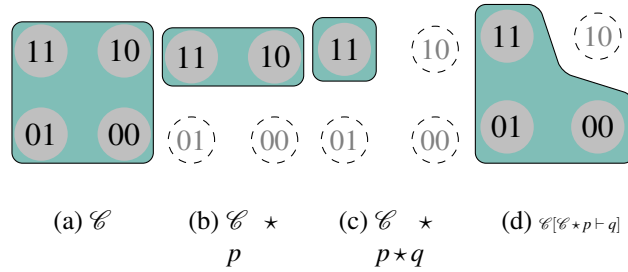


Figure 13

The result is the context shown in Figure 13(d). Importantly, this context is identical to the context obtained by updating the original context with the non-stack conditional $p \rightarrow q$.

Thus the support operation recovers ordinary conditional meanings while preserving the distinction between the actual discourse context and the hypothetical context introduced by the antecedent.

A.2 Popping and Conditional Meanings

After the consequent has been evaluated, the temporary hypothetical context is no longer needed unless it is retained for subsequent discourse, as in cases of modal subordination. Following Isaacs and Rawlins (2008), I therefore assume the operation in (61).

(61) **Definition:** POP

$$\langle \mathcal{C}, \tau \rangle + \text{POP} = \tau$$

if $\tau \neq \langle \rangle$, and $\mathcal{C}$ otherwise.

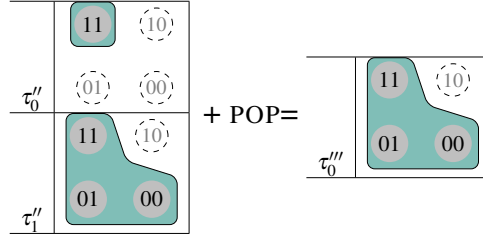


Figure 14 Popping

Applying POP to the macro-context obtained after updating the consequent yields the context shown in Figure 14. The resulting context is identical to the context obtained by updating the original context with the non-stack conditional: $\mathcal{C}[\mathcal{C} \star p \vdash q] = \mathcal{C} \star (p \rightarrow q)$

Thus the stack-based derivation ultimately recovers the familiar conditional interpretation while preserving the distinction between the actual discourse context and the temporary hypothetical context introduced by the antecedent.

A.3 Deriving Consequent Entailment for Unconditional Statements

This appendix illustrates how the present stack-based analysis derives the consequent entailment associated with unconditional statements.

After the consequent proposition has been established in each hypothetical branch, the information percolates to the lower contexts through the MCCP of UPDATE as shown in Figure 9.

Since the temporary hypothetical contexts are no longer needed, they may be removed by a multi-stack version of the POP operation.

(62) **Definition:** MSpop

For any multi-stack $\mathcal{T}$:

$$\mathcal{T} + \text{MSpop} = \langle \tau^{(0)} + \text{POP}, \dots, \tau^{(n)} + \text{POP} \rangle$$

The result is illustrated in Figure 15.

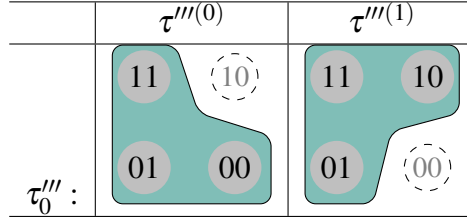


Figure 15 $\tau + \text{whether-or-not}(p) + \text{MSUPDATE}(q) + \text{MSpop}$

To recover the unconditional interpretation, the resulting macro-contexts are merged.

- (63) **Definition:** Stack intersection $\sqcap$
 For any stacks τ and v such that $|\tau| = |v|$,
 $\tau \sqcap v = \chi$
 where $\chi_i = \tau_i \cap v_i$.
- (64) **Definition:** merge
 $\langle \tau^{(0)}, \tau^{(1)} \rangle + \text{merge} = \tau^{(0)} \sqcap \tau^{(1)}$.

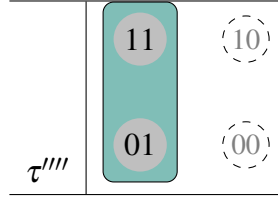


Figure 16 $\langle \tau'''(0), \tau'''(1) \rangle + \text{merge}$

The merge operation collapses the alternative contexts into a single macro-context. As shown in Figure 16, the resulting information state contains only worlds in which q is true. Consequently, the final context is identical to the one obtained by updating the initial context with the simple declarative sentence.

B A formal proof of the equivalence between *whether-or-not*- p , $?q$ and $?q$ given independence

This appendix section gives a formal proof of the intuition that given independence between p and q , an unconditional question *whether or not* p , $?q$ is equivalent $?q$. Let

us borrow the notion of contextual independence defined by Franke (2007, 2009).¹⁵ We consider a context $\mathcal{C} \subseteq \wp(W)$ as the speaker's epistemic context as defined in (24). Let p, q be sentences of $\mathcal{L}$. Following Franke (2009), we introduce the following definitions.

(65) **Definition:** Entailment and Consistency

- a. A context $\mathcal{C}$ entails a sentence p ($\mathcal{C} \Rightarrow p$, in short) in $\mathcal{C}$ if $\mathcal{C} \cap \llbracket p \rrbracket = \mathcal{C}$.
- b. p is consistent ($\Diamond p$, in short) in $\mathcal{C}$ if $\mathcal{C} \cap \llbracket p \rrbracket \neq \emptyset$.

Intuitively, sentences p and q are independent in $\mathcal{C}$ when learning that p or q is true or false does not decide whether the other sentence is true or false:

(66) **Definition:** Contextual Independence

Let $\mathcal{C} \subseteq \wp(W)$ be an epistemic state and p, q sentences. We say that p and q are *contextually independent* in $\mathcal{C}$ if

$$\Diamond x \text{ and } \Diamond y \text{ in } \mathcal{C} \text{ imply } \Diamond(x \wedge y) \text{ in } \mathcal{C},$$

for all $x \in \{p, \neg p\}$ and $y \in \{q, \neg q\}$.

Sano & Hara (2014: 88) rephrase the notion of contextual independence in terms of entailment ' $\Rightarrow$ ' as follows:

(67) **Proposition:**

Let $\mathcal{C} \subseteq \wp(W)$ be an epistemic context and p, q sentences. Then, the following are equivalent:

- (i) p and q are contextually independent in $\mathcal{C}$,
- (ii) if x is consistent in $\mathcal{C}$, then

$$\mathcal{C} \Rightarrow y \text{ is equivalent to } \mathcal{C} \cap \llbracket x \rrbracket \Rightarrow y,$$

for all $x \in \{p, \neg p\}$ and $y \in \{q, \neg q\}$.

Taken together we obtain the following theorem. If we already know that p and q are independent and p is consistent in the context, it follows that $p \rightarrow ?q$ is equivalent to $?q$. That is, $\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q)$ implies $\mathcal{C} \Rightarrow ?q$ and $\mathcal{C} \Rightarrow ?q$ implies $\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q)$.

¹⁵See also Sano & Hara (2014) who provide a dynamic extension of Franke's definition. Note also that in Franke (2009) and Sano & Hara (2014), the notion equivalent to (66) is called "conditional independence".

(68) **Theorem**

Let p and q be independent in $\mathcal{C}$, and let p be consistent with $\mathcal{C}$.
Then, $\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q)$ is equivalent to $\mathcal{C} \Rightarrow ?q$.

(69) *Proof*

- a. $(\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q))$ implies $\mathcal{C} \Rightarrow ?q$

Assume that $\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q)$. That is, $\mathcal{C} \cap \llbracket p \rightarrow ?q \rrbracket = \mathcal{C}$ and $\mathcal{C} \cap \llbracket \neg p \rightarrow ?q \rrbracket = \mathcal{C}$. Thus, $\mathcal{C} \subseteq \{s \in \mathcal{C} \mid \text{for any non-empty } t \subseteq s \text{ if } t \in \llbracket p \rrbracket, \text{ then } t \in \llbracket ?q \rrbracket\}$. We show that $\mathcal{C} \cap \llbracket ?q \rrbracket = \mathcal{C}$. Given independence, it suffices to show that $\mathcal{C} \cap \llbracket p \rrbracket \cap \llbracket ?q \rrbracket = \mathcal{C} \cap \llbracket p \rrbracket$. Since we already know $\mathcal{C} \cap \llbracket p \rrbracket \cap \llbracket ?q \rrbracket \subseteq \mathcal{C} \cap \llbracket p \rrbracket$, we show $\mathcal{C} \cap \llbracket p \rrbracket \subseteq \mathcal{C} \cap \llbracket p \rrbracket \cap \llbracket ?q \rrbracket$. Fix any non-empty $s \in \mathcal{C} \cap \llbracket p \rrbracket$. Then $s \in \llbracket p \rrbracket$. We show that $s \in \mathcal{C} \cap \llbracket p \rrbracket \cap \llbracket ?q \rrbracket$. Since $s \in \mathcal{C}$ and by assumption $s \in \llbracket ?q \rrbracket$, we obtain $s \in \mathcal{C} \cap \llbracket p \rrbracket \cap \llbracket ?q \rrbracket$.

- b. $(\mathcal{C} \Rightarrow ?q)$ implies $\mathcal{C} \Rightarrow (p \rightarrow ?q) \wedge (\neg p \rightarrow ?q)$

Assume that $\mathcal{C} \Rightarrow ?q$. That is, $\mathcal{C} \cap \llbracket ?q \rrbracket = \mathcal{C}$. We show that $\mathcal{C} \cap \llbracket p \rightarrow ?q \rrbracket = \mathcal{C}$ and $\mathcal{C} \cap \llbracket \neg p \rightarrow ?q \rrbracket = \mathcal{C}$. Since we already know $\mathcal{C} \cap \llbracket \neg p \rightarrow ?q \rrbracket \subseteq \mathcal{C}$, we show $\mathcal{C} \subseteq \mathcal{C} \cap \llbracket \neg p \rightarrow ?q \rrbracket$. Fix any non-empty $s \in \mathcal{C}$. We show $s \in \llbracket p \rightarrow ?q \rrbracket$. By assumption, $s \in \llbracket ?q \rrbracket$. Fix any non-empty $t \subset s$ such that $t \in \llbracket p \rrbracket$. Then, there is a non-empty $u \subset t$ such that $u \in \llbracket ?q \rrbracket$, since $u \subset s$. Since s and t are arbitrary, $s \in \llbracket p \rightarrow ?q \rrbracket$. Similarly, we obtain $\mathcal{C} \cap \llbracket \neg p \rightarrow ?q \rrbracket = \mathcal{C}$.